

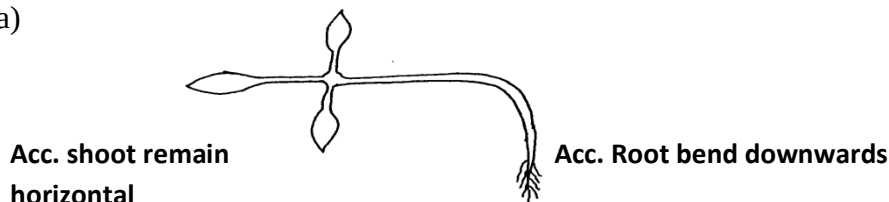


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BIOLOGY PAPER 2
MARKING SCHEME

1. (a) Mastication;/chewing;/grinding; any one (1mk)
- (b) **B** – Gastric juice; (1mk)
C – Intestinal juice;/ succus entericus; (1mk) **reject wrong spelling**
- (c) produces mucus;
 - Enzymes produced in inactive form that is pepsinogen and trypsinogen (2mks)
- (d) Gastrin; (1mk)
- (e) – Enzyme salivary amylase.Ptyalin;
 - Mucin/Mucus;
 - Water; (any two) (2mks)

2. (a)



- (b) (i) Using a **similar** seedling on a clinostat over the same period of time;
- (ii) Radicle and shoot will grow straight/without a curvature due to equal distribution of auxins in the shoot and radical due to rotation
- (c)

Endocrine responses	Nervous responses
Are slow and widespread	Are thick, rapid/fast and localized
Mediated through hormones/chemicals; transported in the blood stream	Mediated through electrical impulses; transmitted through neurones/nerve cells/nerves.

3. (i)

Skeletal (biceps)	Smooth (gut) muscle
• Multinucleated • Striated/stripped • Long muscle fibres • Block/cylindrical	• Uninucleated; • Unstriated; • Short muscle fibres; • Spindle shaped;

(4mks)

- (ii) Intervertebral disc; (1mk)

- (iii) – Acts as a cushion/absorbs shock;



- Reduces friction;
- Flexibility of vertebral column; (3mks)

4. (a) (i)

- To show that energy is released in anaerobic respiration;
- To show that carbon (iv) oxide is produced in anaerobic production; (2mks)

(ii) – Increase in temperature since energy is released;

- Colour of indicator changed to yellow due to acidity/carbon (iv) oxide released changed indicator to yellow; (2mks)

(b) (i) Expel dissolved oxygen; (1mk)

(ii) Prevent entry of air/oxygen into glucose solution; (1mk) **any one**

(c) Use glucose solution without yeast/used to kill yeast cells;

(d) Ethanol production in breweries; bread production in bakeries; **any one** (1mk)

5. (a) Two/2;

X Y; (2mks)

(b)(i) Father $X^H Y$;

Mother $X^H X^h$ (2mks)

(ii) Parental phenotype

Parental genotype

Normal male

$X^H Y$

x

Normal female

$X^H X^h$

√1	♂ ○	♀ ⊙	X^H	X^h	√1
	X^H		$X^H X^{H\sqrt{1}}$	$X^H X^{h\sqrt{1}}$	
	Y		$X^H Y^{\sqrt{1}}$	$X^h Y^{\sqrt{1}}$	

(4mks)

6. (a) **Graph**

(b) (i) $18.5\text{mm} \pm 0.5\text{mm}$; (1mk)

(ii) A number of seeds planted in the same condition/same plot; every week the samples of internode length was measured; total length divided by a number of samples;
max 2 (2mks)

(c) (i) Intermittent; (1mk)

(ii) Sigmoid; (1mk)

(d) (i) Length remains constant/no change in length; presence of rigid exoskeleton/cuticle limit growth; (2mks)

(ii) Length increases; moulting/ecdysis/shedding/growth/expansion of tissues; (2mks)

(e) (i) Apical meristem at the tip of shoot/intercalary meristem/lateral bud at the node divide then cells elongate; (1mk)



(ii) Arthropoda; (1mk)
(rej-anthropoda, arthropod and any wrong spellings)

(iii) Ecdyson hormone; (1mk)
Acc moulting hormone

(iv) $\frac{(24.5-16.5mm)}{(10-7)week}; = \frac{8}{3} = \frac{2.667mm}{week};$ (2mks)

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a) Epidermis has three layers

- i) Cornified layer; Made of dead cells that protects against desiccation/mechanical damages;
- ii) Granular layer; made of living cells that replace worn out cells of cornified layer;
- iii) Malpighian layer; which actively divide to give rise to new epidermal cells
- Has melanin which screens against UV light; cells producing melanin synthesise vitamin D

b) Dermis cells

- (i) Has blood capillaries that supply food/and oxygen/remove excretory products; or arteries that vasolidate when temperatures are low
- (ii) Has sweat glands; that produce sweat which consists of water and salts/sodium chloride /urea and lactic acid; as sweat evaporates, latent heat of vaporization is taken away from the body therefore reducing the body temperatures; under cold conditions, little sweat is produced hence little latent heat of vaporization is taken away from the body therefore conserving heat;
- (iii) Has hair follicles which have hair; which insulates the body against heat loss/hair stand erect to trap air when the temperature is low; hair lie flat to allow heat loss when temperature is high; through this the skin acts as a temperature regulator.
- (iv) Nerve endings; which are sensitive to various stimuli; the skin therefore acts as a sensory organ.
- (v) Sebaceous gland; which produce oily substance sebum which is water repellant/antiseptic; prevents drying and cracking of the skin.
- (vi) Subcutaneous fat; which insulates the body against heat loss.

8 a) Adrenal glands secrete adrenaline into blood stream; cardiac frequency/heart beat increases; while arterioles leading to skeletal muscles dilate; causing more blood to be directed to the tissues; to convey more oxygen; and glucose; and also remove carbon (iv) oxide and other metabolic wastes; liver convert glycogen to glucose; metabolic rate/respiration in skeletal muscles increase; to supply energy; intercostal muscles and diaphragm contract and relax; faster increasing depth of breathing; to supply more oxygen; and remove excess carbon (iv) oxide from the lungs (14mks)

(b) (i) Reflex action; (1mk)
(ii) Conditioned reflex; (1mk)

REFLEX ACTION	CONDITIONED REFLEX
Single stimulus bring about response	Original stimulus associated with a new stimulus to bring response

Independent of experiences	Requires experience
Sensory component remains the same	Primary sensory component replaced by secondary component
Reflex simple	Reflex modified

6.

